

LEVEL 2: 2–5 SECOND RAPID ELIMINATION

SINGLE-STEP VALUE PUTTING • MANDATORY PAIR FINDERS • SUB-5s OPTION DESTROYERS

Contains 20 Master Exam Questions across Reasoning, Maths, English & GK with Examiner Trap Justifications and Sub-5-Second Execution Keys

LEVEL 2 SPEED PHILOSOPHY & JUSTIFICATION

Why 2–5 Seconds is 100% Possible: Level-2 questions involve single-step mathematical substitutions (e.g. angle putting $\theta=45^\circ$, fraction ladder shifts, or locating Noun-Pronoun mandatory pairs in PQRS). Rather than solving long analytical proofs or parsing entire paragraphs, you identify the **structural anchor** and eliminate 3 options in one swift mental stroke.

Q1. [Maths: Trigonometry Tan-Cot 45 Pin]

Speed Target: 3s

Question: Find the value of: $[(1 + \tan\theta + \sec\theta)(1 + \cot\theta - \csc\theta)]$

Options: A) 0 B) 1 C) 2 D) -1

Examiner's Trap: Expands tan, cot, sec, csc into sin and cos, leading to a 10-line algebraic simplification.

Smartest Trick Justification: "Tan-Cot loves 45. When tan/cot/sec/csc appear, put $\theta = 45^\circ$ ($\tan 45=1$, $\cot 45=1$, $\sec 45=\sqrt{2}$, $\csc 45=\sqrt{2}$)."

Step-by-Step Speed Kill: Plug $\theta=45^\circ$: $(1 + 1 + \sqrt{2})(1 + 1 - \sqrt{2}) = (2 + \sqrt{2})(2 - \sqrt{2}) = 2^2 - (\sqrt{2})^2 = 4 - 2 = 2$. → Answer: C (2)

Q2. [English: PQRS Mandatory Noun-Pronoun Hook]

Speed Target: 3-4s

Question: Arrange the sentences: P: He spent his life researching gravity. Q: Isaac Newton was born in 1643. R: This law changed physics. S: He published the Principia in 1687.

Options: A) PQRS B) QPSR C) RQPS D) SPRQ

Examiner's Trap: Forces reading and comprehending all 4 sentences in chronological sequence (45 seconds).

Smartest Trick Justification: "Find the Noun before the Pronoun. Sentence Q introduces 'Isaac Newton'. Sentence P refers to 'He'. Q MUST come before P."

Step-by-Step Speed Kill: Q introduces Newton → Only B starts with Q. Solved in 3s without reading R and S! → Answer: B (QPSR)

Q3. [Reasoning: Paper Folding Single-Notch Filter]

Speed Target: 3s

Question: A square sheet is folded twice into a quadrant, and a single triangular cut is made pointing LEFT in the bottom-right corner. When unfolded, what must be true?

Options: A) All 4 triangles point left.
B) Adjacent triangles point in opposite directions (mirror reflection).
C) All triangles point upwards.
D) Triangles rotate 90 degrees.

Examiner's Trap: Buries candidates in complex 16-cut mental visualization.

Smartest Trick Justification: "Track one sharp corner, ignore the rest. Unfolding across a vertical fold flips left-pointing to right-pointing."

Step-by-Step Speed Kill: Bottom-right points left → bottom-left MUST point right (mirror). Eliminate all options violating this. → Answer: B

Q4. [Maths: Cyclic Symmetric Algebra]

Speed Target: 3s

Question: If $x + y + z = 6$ and $xy + yz + zx = 11$, find the value of $x^3 + y^3 + z^3 - 3xyz$ when $x=1$, $y=2$, $z=3$.

Options: A) 18 B) 24 C) 36 D) 54

Examiner's Trap: Forces formula: $(x+y+z)[(x+y+z)^2 - 3(xy+yz+zx)]$.

Smartest Trick Justification: "Symmetric variables satisfy natural integers: $1+2+3 = 6$, $(1 \times 2 + 2 \times 3 + 3 \times 1) = 2+6+3 = 11$. $x=1$, $y=2$, $z=3$."

Step-by-Step Speed Kill: Formula = $6 \times [6^2 - 3(11)] = 6 \times (36 - 33) = 6 \times 3 = 18$. → Answer: A (18)

Q5. [GK: Extreme Words Statement Filter]

Speed Target: 2-3s

Question: Which of the following statements regarding the Monetary Policy Committee (MPC) is correct?

Options: A) The Governor of RBI ALWAYS makes decisions solely without voting.
 B) The MPC consists of 6 members and meets at least 4 times a year.
 C) Government nominees COMPLETELY control interest rates.
 D) Inflation targets NEVER change under any circumstances.

Examiner's Trap: Sets up long multi-line economic policy statements to drain reading time.

Smartest Trick Justification: "Absolutes lie, qualifiers fly. Statements with 'Always', 'Solely', 'Completely', 'Never' are 90% false."

Step-by-Step Speed Kill: Options A, C, D contain 'ALWAYS', 'COMPLETELY', 'NEVER'. Eliminate all three in 2s → Option B is correct.

→ Answer: B

Q6. [English: Active/Passive Continuous Tense]

Speed Target: 2s

Question: Convert to passive: 'The municipal corporation is constructing a new bridge over the river.'

Options: A) A new bridge was constructed by the municipal corporation.
 B) A new bridge has been constructed by the municipal corporation.
 C) A new bridge is being constructed by the municipal corporation.
 D) A new bridge was being constructed by the municipal corporation.

Examiner's Trap: Swaps continuous helping verbs with past and perfect forms.

Smartest Trick Justification: "Present continuous (is/are + V-ing) MUST become is/are BEING + V3."

Step-by-Step Speed Kill: Scan for 'is being constructed' → Only Option C matches. → Answer: C

Q7. [Reasoning: Letter Coding Position Shift]

Speed Target: 3s

Question: In a certain code language, 'BIRD' is coded as 'DKTF'. How is 'SWAN' coded in that language?

Options: A) UYCP B) UXBO C) VZCP D) TYDQ

Examiner's Trap: Candidates write long alphabet tables and map all 4 letters.

Smartest Trick Justification: "Check first and last letter shifts: B(+2)=D, D(+2)=F → Rule is +2 on all letters."

Step-by-Step Speed Kill: S(+2) = U; N(+2) = P. Must start with U and end with P → Option A (UYCP). → Answer: A (UYCP)

Q8. [Maths: Digital Root Multiplication Check]

Speed Target: 3-4s

Question: Find the exact product: $513 \times 426 = ?$

Options: A) 218,538 B) 218,532 C) 218,540 D) 218,528

Examiner's Trap: Forces massive 3-digit multiplication by hand.

Smartest Trick Justification: "Digital Root (Mod 9) Radar: Invariant under multiplication."

Step-by-Step Speed Kill: DR(513) = $5+1+3 = 9$ (0). Any number $\times 0$ has Digital Root 9 (0). Check options: Option A ($2+1+8+5+3+8 = 27 \rightarrow 9$). Only A matches! → Answer: A (218,538)

Q9. [English: Spotting Error Subject-Verb]

Speed Target: 2s

Question: Spot the error: 'Either the manager or his assistants has submitted the quarterly report.'

Options: A) Either the manager B) or his assistants C) has submitted D) the quarterly report

Examiner's Trap: Places singular 'manager' first to mask plural verb requirement.

Smartest Trick Justification: "In 'Either...or', the verb ALWAYS agrees with the CLOSEST subject."

Step-by-Step Speed Kill: Closest subject is plural 'assistants' → requires 'have submitted' (not 'has'). Error is C. → Answer: C

Q10. [Reasoning: Number Analogy Smallest-to-Largest]

Speed Target: 4s

Question: Select the related number triplet matching: (8, 14, 220).

Options: A) (5, 9, 120) B) (6, 11, 170) C) (7, 12, 190) D) (4, 10, 140)

Examiner's Trap: Sets up obscure arithmetic polynomial operations.

Smartest Trick Justification: "Operate on smallest two (A & B) to form largest (C): $(A + B) \times k$."

Step-by-Step Speed Kill: Triplet: $(8 + 14) = 22 \times 10 = 220$. Multiplier is 10. Check Option B: $(6 + 11) = 17 \times 10 = 170$. Matches! → Answer: B ((6, 11, 170))

Q11. [GK: Dance Gharana Exponent]

Speed Target: 2s

Question: Raja and Radha Reddy are world-renowned exponents of which classical dance?

Options: A) Kathak B) Kuchipudi C) Bharatanatyam D) Kathakali

Examiner's Trap: Tests specific state dance pairs.

Smartest Trick Justification: "Reddy / Raja & Radha = Kuchipudi (Andhra Pradesh)."

Step-by-Step Speed Kill: Scan surname 'Reddy' → Kuchipudi in 1 second. → Answer: B (Kuchipudi)

Q12. [Maths: Profit & Loss Proportionality]

Speed Target: 4s

Question: A shopkeeper marks his goods 40% above CP and allows a 20% discount. What is his profit percentage?

Options: A) 10% B) 12% C) 15% D) 20%

Examiner's Trap: Leads students into multi-step $CP=100 \rightarrow MP=140 \rightarrow SP=140 \times 0.8$ calculation.

Smartest Trick Justification: "Successive Formula: $\text{Profit}\% = x - y - (xy/100)$ where $x=\text{markup}$, $y=\text{discount}$."

Step-by-Step Speed Kill: $\text{Profit}\% = 40 - 20 - (40 \times 20/100) = 20 - 8 = 12\%$. Solved in 3s. → Answer: B (12%)

Q13. [English: Direct/Indirect Modal Backshift]

Speed Target: 2s

Question: Convert to indirect speech: The doctor said, 'You can resume exercise next week.'

Options: A) The doctor said that I can resume exercise next week.
B) The doctor said that I could resume exercise the following week.
C) The doctor said that I should resume exercise next week.
D) The doctor said that I might resume exercise next week.

Examiner's Trap: Tests both modal backshift (can → could) and time adverbial shift (next week → following week).

Smartest Trick Justification: "Said + Can MUST become COULD; 'next week' becomes 'the following week'."

Step-by-Step Speed Kill: Scan for 'could' and 'the following week' → Option B. → Answer: B

Q14. [Reasoning: Clock Angle Shortcut]

Speed Target: 4s

Question: Find the angle between the hour hand and minute hand at 4:20.

Options: A) 0° B) 10° C) 15° D) 20°

Examiner's Trap: Forces drawing clock dials and computing (30H - 5.5M) manually.

Smartest Trick Justification: "Clock Ratio Shortcut: When Time ratio is 1:5 (4:20 → 1:5), Angle = Minutes / 2."

Step-by-Step Speed Kill: Angle = $20 / 2 = 10^\circ$. Solved in 2s! → Answer: B (10°)

Q15. [Maths: Quadratic Roots Symmetry]

Speed Target: 3s

Question: If α and β are roots of $x^2 - 7x + 12 = 0$, find the value of $(\alpha^2 + \beta^2)$.

Options: A) 25 B) 37 C) 49 D) 13

Examiner's Trap: Students solve quadratic formula: $x = 3, 4$, then square $9 + 16$.

Smartest Trick Justification: "Direct Identity: $\alpha^2 + \beta^2 = (\text{Sum})^2 - 2(\text{Product}) = b^2 - 2c$."

Step-by-Step Speed Kill: Sum = 7, Product = 12. $\alpha^2 + \beta^2 = 7^2 - 2(12) = 49 - 24 = 25$. → Answer: A (25)

Q16. [GK: Constitutional Article Ordinance]

Speed Target: 2s

Question: Under which Article can the Governor of a State promulgate an Ordinance?

Options: A) Article 123 B) Article 213 C) Article 214 D) Article 163

Examiner's Trap: Confuses President's Ordinance (123) with Governor's Ordinance (213).

Smartest Trick Justification: "Rearrange Digits Rule: President (Art 123) → Move 1 to middle: Governor (Art 213)."

Step-by-Step Speed Kill: 123 → 213 → Article 213. Done in 1s. → Answer: B (Article 213)

Q17. [English: Vocabulary Root Word Lock]

Speed Target: 2s

Question: What is the meaning of 'Somnambulist'?

Options: A) One who talks in sleep B) One who walks in sleep C) One who hates sleep D) One who loves sleep

Examiner's Trap: Presents confusing sleep disorders (somniloquy vs somnambulism).

Smartest Trick Justification: "Root Words: Somn = Sleep, Ambul = Walk (like ambulance / amble)."

Step-by-Step Speed Kill: Somn + Ambul = Sleep + Walk → One who walks in sleep. → Answer: B

Q18. [Reasoning: Number Series Difference Pattern]

Speed Target: 4s

Question: Find the next number in series: 4, 9, 19, 39, 79, ?

Options: A) 149 B) 159 C) 169 D) 179

Examiner's Trap: Candidates calculate second-order difference triangles.

Smartest Trick Justification: "Recognize pattern: $(\times 2 + 1)$ across all terms."

Step-by-Step Speed Kill: $79 \times 2 + 1 = 158 + 1 = 159$. Solved in 3s. → Answer: B (159)

Q19. [Maths: Remainder Theorem Shortcut]

Speed Target: 3s

Question: Find the remainder when $(67^{67} + 67)$ is divided by 68.

Options: A) 1 B) 66 C) 67 D) 0

Examiner's Trap: Forces binomial expansion of 67^{67} .

Smartest Trick Justification: "Modulo -1 Rule: $67 \equiv -1 \pmod{68}$. $(-1)^{\text{odd}} = -1$."

Step-by-Step Speed Kill: Remainder = $(-1 + 67) = 66$. Solved in 2s! → Answer: B (66)

Q20. [English: Inversion with Seldom / Hardly]

Speed Target: 2s

Question: Spot the error: 'Seldom we have seen such a breathtaking natural landscape.'

Options: A) Seldom we have B) seen such C) a breathtaking D) natural landscape

Examiner's Trap: Masks negative inversion requirement at the beginning of sentences.

Smartest Trick Justification: "When a sentence begins with Seldom/Hardly/Never, auxiliary verb **MUST** precede subject (Inversion)."

Step-by-Step Speed Kill: 'Seldom we have' must be 'Seldom have we'. Error is A. → Answer: A